



SIMATS SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105



CSA0506 – Database Management System

The Importance of Database Management Systems in Modern Life

Supervisor:

Dr.R.Hemavathi

Presented by:

Harithaa S



INTRODUCTION

- A **database** is a structured collection of related data that is organized for easy storage, retrieval, and management.
- It stores information in tables, making it simple to search, update, and maintain records efficiently.
- A **Database Management System (DBMS)** is software that enables users to create, organize, manage, and control databases. It provides functions such as adding, editing, deleting, and retrieving data while ensuring data security, accuracy, and consistency. \
- DBMS also supports multiple users accessing the same database at the same time.
- It is widely used in real-world applications such as hospital patient records, online shopping websites, school management systems, banking applications, libraries, and airline reservation systems.



OBJECTIVES AND FEATURES

Objectives

- Organize large amounts of information.
- Prevent duplicate data.
- Improve data accuracy.
- Protect confidential information.
- Support multiple users at the same time.

Key Characteristics

- High reliability
- Data independence
- Easy maintenance
- Efficient data retrieval
- Access control



APPLICATIONS



Healthcare

- Patient registration
- Lab reports
- Medical history

Education

- Student admission
- Examination records
- Library management

Finance

- Account management
- Digital transactions
- Loan processing

Retail

- Inventory management
- Billing system
- Customer purchase history

Travel

- Hotel reservations
- Flight booking
- Passenger information

Entertainment

- Streaming subscriptions
- User preferences
- Watch history



EXAMPLE –ONLINE SHOPPING SYSTEM



How DBMS Works

Step 1

- User logs into the application.

Step 2

- Searches for a restaurant or food item.

Step 3

- DBMS checks restaurant availability, menu, and prices.

Step 4

- Customer places the order.

Step 5

- Database stores:
 - Customer details
 - Delivery address
 - Payment status
 - Order details

Step 6

- Order is assigned to a delivery partner.
- Customer receives live tracking and confirmation



Advantages and Disadvantages

Advantages:

- 1. Reduces Data Redundancy:** Eliminates duplicate data and saves storage space.
- 2. Improves Data Security:** Protects sensitive information using authentication and access control.
- 3. Ensures Data Consistency:** Keeps data accurate and up to date across the database.
- 4. Supports Multiple Users:** Allows many users to access and update data at the same time.

Disadvantages:

High Cost: Installation, maintenance, and licensing can be expensive.

- 1. Complexity:** Requires skilled professionals to manage and maintain the database.
- 2. Performance Overhead:** Can be slower than file systems for small applications.
- 3. Risk of System Failure:** If the database server fails, data access may be interrupted until recovery.



CONCLUSION

1. DBMS enables efficient storage, organization, and management of data.
2. It minimizes duplicate data and maintains data accuracy and consistency.
3. It enhances data security by safeguarding confidential information.
4. It supports concurrent access, allowing multiple users to work with the same database at the same time.
5. It provides quick data retrieval and simplifies data modification and maintenance.
6. DBMS is extensively used in sectors such as banking, healthcare, education, retail, and e-commerce.
7. Emerging technologies, including cloud computing and artificial intelligence, continue to improve the capabilities of DBMS.
8. In conclusion, DBMS plays a vital role in ensuring secure, reliable, and efficient data management across various industries.